
How Do World Action Models Use Their Futures?

Evidence from Mechanistic Interventions

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Abstract

World action models generate future observations alongside robot actions, but co-generation does not show that the future affects the action. We use a two-stage intervention. Stage 1 starts with two normal rollouts from the same simulator state. We insert the predicted future from one rollout (the *donor*) into the other (the *recipient*), hold the recipient’s observation, instruction, random draws, and solver fixed, and ask whether its recomputed action moves toward the donor action, which the model never sees. Across 22 states and six RoboLab tasks, inserted donor futures redirect Cosmos 3 actions and executed endpoints almost completely toward the donor rollout. Cosmos Policy shows a smaller effect at the first decision state of all ten LIBERO-Long tasks. Matched Gaussian, shuffled, self-recomputation, and alternative natural-future controls rule out perturbation magnitude and generic future replacement as complete explanations. To determine what specific future information shapes action, we separately test predicted wrist-camera and primary-camera views, future proprioception, and robot- versus object-pixel content. Cosmos Policy responds primarily to camera-visible robot motion, whereas isolated robot or object pixels do not reproduce Cosmos 3’s whole-future effect. In Stage 2, we keep the donor future inserted but replace the future tokens’ attention keys and values with those from the recipient’s original-future run. This removes 83–88% of Cosmos 3 steering in every tested state, showing that this pathway carries most of the future’s influence. Together, the interventions identify different mechanisms: an early, wrist-camera-dominated future signal in Cosmos Policy and, in Cosmos 3, a whole-future signal not reproduced by isolated robot or object pixels, with most of its causal influence carried through future-token K/V.

1 Introduction

Many world action models (WAMs) generate future observations alongside robot actions [8, 14]. Joint generation, however, does not imply that the action computation uses the generated future. The policy could map the current observation and instruction directly to an action while producing a plausible future as an auxiliary output.

Future representations could serve many roles: they may be auxiliary outputs, a general computational workspace, a content-specific input such as a prospective robot pose used for inverse dynamics [1, 7], or a representation of consequences used to evaluate candidate actions [2, 3]. These possibilities are neither exhaustive nor mutually exclusive. RIFT provides the closest prior causal evidence [22]: masking future reads or editing values at future positions changes trajectories and task success across several WAMs. Such destructive interventions establish dependence on the future channel, but damage can change an action in any direction. They do not show that one coherent alternative future causes the action associated with that future.

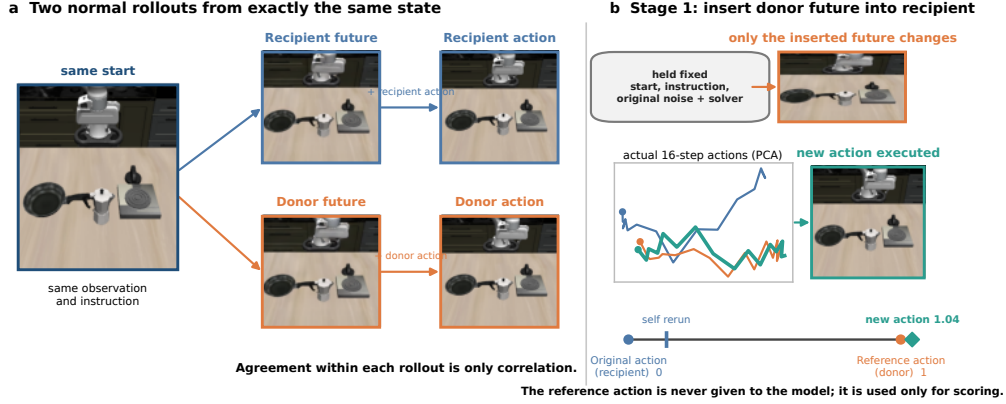


Figure 1: Stage-1 future insertion on a held-out LIBERO trace. Blue, orange, and green denote the recipient, donor, and recomputed action. Action curves are projected to two dimensions for display; inference uses the full action vectors.

We therefore run a simple causal test (Figure 1): we take the future predicted in one normal rollout and insert it into another normal rollout that starts from exactly the same saved state. We call the rollout receiving the future the *recipient* and the rollout supplying it the *donor*. We then recompute the recipient’s action while keeping everything else fixed. The model never sees the donor’s action; we use that action only after inference as a reference. If the recipient’s new action moves toward the donor’s action, then the content of the inserted future—rather than merely activity at future-token positions—has a directional causal effect on behavior.

We use two distinct intervention stages. In *Stage 1*, we patch the future produced by the donor rollout into the recipient run and ask *whether* changing that future’s content steers behavior. In *Stage 2*, we keep the donor future in the recipient run but replace only the donor-run future-token keys and values with those recorded during a recipient self-future run. This K/V pathway patch asks *how* the Stage-1 effect reaches the action. It is a nested mechanism test, not another name for patching the donor’s predicted future into the recipient.

We study Cosmos Policy’s direct-policy mode and the diffusion-generator component of Cosmos3-Nano-Policy-DROID. Neither evaluated inference procedure generates and scores multiple candidate actions before acting; Cosmos Policy’s separate best-of- N mode is outside our study. Thus, content-specific future conditioning would not by itself demonstrate outcome comparison or planning.

We ask whether alternative futures steer actions and executed endpoints, which visual content is sufficient, and which internal pathway carries the effect. Cosmos 3 provides the strongest chain: steering generalizes across 22 states, isolated robot or object pixels do not reproduce it, and future-token K/V replacement suppresses action and endpoint steering in every state. Cosmos Policy supplies a cross-generation comparison whose effect is concentrated at early states and is most consistent with camera-visible robot motion.

2 Related Work

Prediction and control. Some visual robot policies generate future frames and infer actions from them; others condition action inference on predicted video representations [1, 7]. Joint video–action models instead learn future observations and controls in one model [6, 8, 10, 14, 20]. Their training objective does not determine whether a particular generated future is used at inference. UVA, for example, can omit video generation at test time [10]. Explicit visual planners answer a different question by sampling alternatives, predicting their outcomes, and comparing them with a goal-dependent score [2, 3].

Recent WAM studies separate predictive training from the use of predictions during action generation. Fast-WAM removes explicit test-time rollout while retaining competitive performance in its evaluations [21]; Faster-WAM retains a lower-cost future-conditioning route and reports improved

robustness [24]. AGRA finds that visually plausible foresight need not support reliable action inference in its studied architecture [16]. RIFT directly establishes dependence on future-position values and their organization [22]. Our contribution is complementary: RIFT asks whether behavior depends on the future interface, whereas our intervention asks whether inserting a coherent alternative future predicts the direction of behavior.

Mechanistic interventions. Activation replacement tests whether an internal representation affects an output rather than merely being decodable from it [12, 18]. Path patching extends this logic to specified computational routes [5]. Conclusions can depend on the counterfactual pair and metric [23], while multi-component effects need not add when patched and unpatched routes interact [17]. We therefore treat K/V replacement as evidence for an active pathway, not a complete circuit decomposition.

3 Stage 1: Patching the Donor Future into the Recipient Run

Stage 1 asks whether the specific content of a model-generated future helps determine the action, or whether the future is merely an accompanying prediction.

3.1 Method

The Stage-1 procedure has four steps:

1. **Generate two possibilities.** Starting from the same saved simulator state, run the model twice to obtain two different predicted futures and their corresponding actions. The rollout that will receive the future insertion is the *recipient*; the rollout supplying the future and reference action is the *donor*.
2. **Insert one future into the other run.** Restore the recipient to the same starting state and replace only its predicted future with the donor’s predicted future.
3. **Recompute the action.** Run the recipient again while keeping its observation, instruction, random draws, solver, and all action coordinates fixed. The model never receives the donor’s action.
4. **Measure the direction of change.** After inference, compare the recomputed recipient action with the original recipient and donor actions. Movement toward the donor action shows that the inserted future’s specific content steered behavior.

The controls test whether this movement can instead be explained by recomputation, perturbation size, or inserting any alternative future. We next give the formal implementation and then report the behavioral, content, and scope results.

3.1.1 Formal recipient–donor design and outcome

Let saved simulator state S admit two unmodified rollouts: recipient R and donor D , with future representations F_R, F_D , action chunks $\mathbf{a}_R, \mathbf{a}_D$, and endpoints $\mathbf{e}_R, \mathbf{e}_D$ obtained by executing one action chunk. The recipient is the rollout whose action is recomputed; the donor supplies the alternative future and defines the scoring direction. We recompute R from the identical state while retaining its observation, instruction, action noise, future-noise path, solver, and denoising schedule, and replace only F_R with F_D . We also test the reciprocal intervention, so each rollout serves once in each role.

For an action or endpoint $\mathbf{x} \in \{\mathbf{a}, \mathbf{e}\}$, normalized donor projection is

$$\pi_x(\hat{\mathbf{x}}; R, D) = \frac{(\hat{\mathbf{x}} - \mathbf{x}_R)^\top (\mathbf{x}_D - \mathbf{x}_R)}{\|\mathbf{x}_D - \mathbf{x}_R\|_2^2}. \quad (1)$$

The recipient is 0, the donor is 1, and 0.5 is halfway between them along the donor direction; negative values move away and values above 1 overshoot. Because the dot product retains only this signed direction, a large change perpendicular to the donor receives little projection. The primary steering contrast subtracts the corresponding self-future recomputation:

$$\Delta_x = \pi_x(\hat{\mathbf{x}}_{\text{donor}}; R, D) - \pi_x(\hat{\mathbf{x}}_{\text{self}}; R, D). \quad (2)$$

115 The self condition reruns the same intervention machinery with the recipient’s own future, so subtraction removes any displacement due merely to recomputation. Pairs are selected using native
116 action and endpoint separation, never intervention outcomes. Every endpoint arm is executed after
117 restoring the exact simulator state.
118

119 3.1.2 How the donor future is patched into the recipient run

120 In both models, we overwrite only the recipient’s future coordinates with the donor’s values at every
121 denoising step; the recipient observation, instruction, random draws, solver schedule, and action
122 coordinates remain fixed. For Cosmos Policy, we replace the clean future-proprioception and two
123 future-camera latents at each solver evaluation. For Cosmos 3, we replace the future-video frames
124 before both guidance branches and again at the sampler boundary, using either a predicted donor
125 latent or a video obtained by executing the donor action. Exact tensor layouts, schedules, and
126 sampler equations are in Appendix A.2.

127 3.1.3 Models, populations, and controls

128 We evaluate Cosmos-Policy-LIBERO-Predict2-2B on LIBERO-Long: ten tasks at the first
129 query and after three and six open-loop chunks, for 30 registered states [8, 11]. We also evaluate the
130 16B Cosmos3-Nano-Policy-DROID checkpoint on six RoboLab tasks: 24 state clusters were
131 frozen without intervention outcomes and 22 completed [14, 15, 19]. Native recipient–donor pairs
132 are chosen for action and executed-endpoint separation before intervention; reciprocal directions
133 and repeated noise draws are averaged within state.

134 Controls include self-future recomputation, norm- and distance-matched Gaussian targets, prespeci-
135 fied natural futures, within-task shuffled futures, and bitwise no-op checks. These distinguish recom-
136 putation, perturbation magnitude, generic valid-future effects, and pair-specific content. Executed-
137 future conditions additionally test whether steering survives when the donor source is physically
138 realized. Full checkpoint, sampling, selection, restoration, and control specifications are in Ap-
139 pendix A.1–A.5.

140 3.1.4 Content interventions

141 We isolate content in complementary ways. In Cosmos Policy, first-query interventions replace the
142 wrist-camera future, primary-camera future, or future proprioception separately. A second study
143 crosses binary robot and object states in re-rendered videos at 20 held-out states. A decoder checks
144 whether the intended cell remains visible. A natural-pair screen separately searches policy-generated
145 continuations for robot-divergent/object-matched and object-divergent/robot-matched futures; only
146 four states meet its criteria, below the planned minimum of ten.

147 For Cosmos 3, a separately frozen factor study selected 12 population states round-robin by native
148 object-position separation. Ten completed across all six tasks. Starting from the recipient video, an
149 object-priority disjoint mask copies donor object pixels, donor robot pixels outside the object mask,
150 both, or neither. This 2×2 design tests isolated pixel factors; it does not guarantee an in-distribution
151 video or isolate a latent semantic variable.

152 Stage-1 statistical procedures, including independent-unit aggregation, bootstrap intervals, multi-
153 plicity correction, and the prespecified smallest effect of interest, are detailed in Appendix A.6.

154 3.2 Results

155 3.2.1 Do coherent futures steer behavior?

156 The method succeeds if three things happen: inserting a donor future moves behavior toward that
157 donor, the shift survives simulator execution, and the paired donor outperforms non-donor controls.
158 Table 1 puts the method and every applicable control on one scale, organized by model, future
159 source, and outcome. Figure 2 separates the two statistical questions. Panel a reports donor-minus-
160 self steering, for which positive values indicate movement toward the paired donor and 1 denotes
161 the full native recipient-to-donor displacement. Panel b reports donor-minus-control contrasts, for
162 which positive values mean that the paired donor steers behavior more strongly than the stated
163 control.

Table 1: Stage-1 projections split by model. Zero denotes recipient self-future behavior and one the paired donor. Bold marks the largest estimate within each source–outcome block. Controls are reconstructed within independent unit; Cosmos 3 endpoint controls are secondary.

(a) Cosmos 3			(b) Cosmos Policy		
Source/outcome	Future	Projection [95% CI]	Outcome	Future	Projection [95% CI]
Pred. action	Self	0.000 (ref.)	Action	Self	0.000 (ref.)
	Natural	0.559 [0.465, 0.651]		Gaussian	0.001 [−0.004, 0.006]
	Paired donor	0.992 [0.972, 1.010]		Natural	0.395 [0.269, 0.553]
Exec. action	Self	0.000 (ref.)		Shuffled	0.244 [0.152, 0.347]
	Gaussian	0.041 [0.003, 0.076]	Paired donor 0.499 [0.336, 0.679]		
	Natural	0.591 [0.491, 0.687]	Endpoint	Self	0.000 (ref.)
	Paired donor	0.983 [0.890, 1.064]		Gaussian	0.003 [−0.003, 0.008]
Exec. endpoint	Self	0.000 (ref.)		Natural	0.442 [0.311, 0.598]
	Gaussian	−0.041 [−0.168, 0.054]		Shuffled	0.275 [0.185, 0.375]
	Natural	0.514 [0.387, 0.642]		Paired donor	0.552 [0.387, 0.728]
	Paired donor	1.003 [0.950, 1.050]			

Cosmos 3. Across 22 completed states and all six tasks, a predicted paired-donor future produces action steering of 0.992 ([0.972, 1.010]; 22/22 states positive), compared with 0.559 for the natural-future control (Table 1). Executed paired-donor videos produce action steering of 0.983, compared with 0.591 for the natural control and 0.041 for the geometry-matched Gaussian control. Executing the paired-donor-conditioned action yields endpoint steering of 1.003 ([0.950, 1.050]; 22/22). The corresponding endpoint projections are 0.514 ([0.387, 0.642]) for the natural control and −0.041 ([−0.168, 0.054]) for the matched Gaussian. In paired secondary analyses, the donor therefore exceeds the natural endpoint control by 0.488 ([0.348, 0.631]; 21/22 states positive) and the Gaussian endpoint control by 1.043 ([0.946, 1.168]; 22/22). Thus, the result is not explained by recomputation, an equally large unstructured displacement, or merely supplying a different valid continuation. Endpoint projection shows that the donor-directed component survives simulator execution; it does not imply equality of the full endpoint vectors. Task-balanced analyses preserve these conclusions.

Cosmos Policy. At the first query, the paired donor produces action steering of 0.499 and endpoint steering of 0.552, positive in all ten tasks (Table 1). The natural control reaches 0.395 and 0.442, respectively, while the shuffled control reaches 0.244 and 0.275 and the matched Gaussian remains near zero. The natural future is the hardest control: the paired donor exceeds it by 0.103 ([0.048, 0.161]; 8/10) for actions and 0.110 ([0.054, 0.166]; 8/10) for endpoints. This rules out the weak explanation that any different tokenizer-consistent future causes the same movement. Appendix Figure 6 examines how this effect varies across state cohorts and future components.

No screened branch pair in either policy retained a stable success/failure label across continuation seeds. We therefore interpret these results as donor-directed changes in action and one-chunk endpoint, not changes in closed-loop task completion.

3.2.2 What future content causes steering?

The content that drives steering differs across checkpoints. Cosmos Policy responds mainly to camera-visible robot motion at early states, whereas neither isolated pixel factor recreates Cosmos 3’s whole-future effect.

Cosmos Policy. Wrist-future action steering is 0.435 ([0.284, 0.605]), compared with 0.101 ([0.057, 0.144]) for the primary camera and 0.001 ([−0.004, 0.008]) for future proprioception (Appendix Figure 6b). In the 20-state 2×2 study, decoded futures identify the intended cell with 92.5% primary-camera and 98.75% wrist-camera accuracy. The intended object manipulation is therefore visible in the rendered futures, yet the object-state main effect is 0.00015 ([−0.00012, 0.00049]) for action and 0.00018 ([−0.00040, 0.00102]) for the executed goal-state endpoint. The available analysis does not report the robot-state main effect. Four exploratory natural pairs provide a complementary comparison: robot-motion futures yield action steering of 0.291 ([0.060, 0.521]) and robot-endpoint steering of 0.315 ([0.059, 0.571]), whereas the object-only effects are 0.021 for both action and goal endpoint, with intervals crossing zero. At these sampled early states, the pattern is most consistent with an inverse-dynamics-like role for camera-visible robot motion. It does not show that Cosmos Policy generally ignores object consequences.

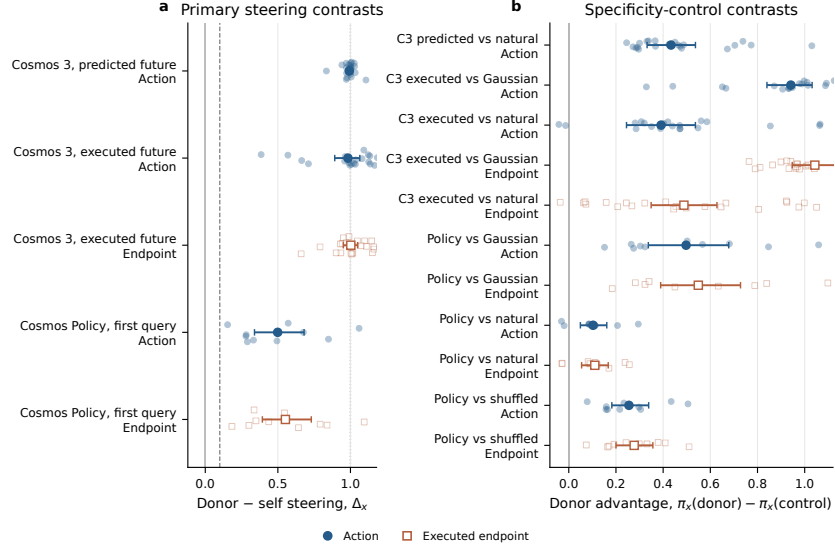


Figure 2: Stage-1 directional steering (a) and donor-minus-control contrasts (b). Small markers are states for Cosmos 3 and tasks for Cosmos Policy; large markers and whiskers are means and 95% bootstrap intervals. Blue denotes actions and orange denotes endpoints.

Cosmos 3. Across ten factor-study states and all six tasks, every isolated-factor interval lies inside the prespecified $[-0.10, 0.10]$ equivalence region (Table 2; Figure 3). Panels a and b show all four cells of the 2×2 hybrid intervention: swapping robot pixels, object pixels, or both changes neither the action nor the executed endpoint appreciably. Panel c places the factor effects beside the whole-future intervention on the same axis. Whole-future steering is approximately one unit, whereas robot-pixel effects are 0.0022 for actions and 0.0102 for endpoints, and object-pixel effects are 0.0061 and -0.0001 . Neither isolated factor therefore recreates a meaningful fraction of the whole-future effect. The useful signal may be distributed across the image, require robot-object coherence, depend on camera or temporal structure, or interact nonlinearly with masking and tokenization. Hybrid videos may also be off distribution. The experiment establishes insufficiency of each isolated pixel factor, not irrelevance of robot or object information and not joint necessity.

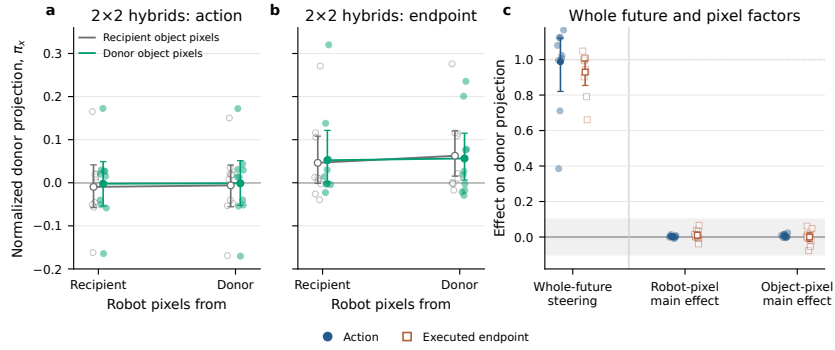


Figure 3: Cosmos 3 robot/object pixel factorization. Panels (a)–(b) show the four hybrid-video cells; panel (c) shows robot/object main effects and the whole-future benchmark. Small markers are states, large markers and whiskers are means and 95% bootstrap intervals, and the gray band is $[-0.10, 0.10]$.

Table 2: Cosmos 3 isolated-content interventions across 10 states from six tasks. Entries are mean donor-directed projections with 95% state-bootstrap intervals; 0 denotes the recipient and 1 the paired donor. Robot and object masks are disjoint. The prespecified practical-equivalence region is $[-0.10, 0.10]$.

Donor pixels inserted	Action projection [95% CI]	Endpoint projection [95% CI]
Robot only	0.0022 $[-0.0016, 0.0060]$	0.0102 $[-0.0067, 0.0275]$
Object only	0.0061 $[0.0009, 0.0119]$	-0.0001 $[-0.0254, 0.0239]$

3.2.3 Where is the Cosmos Policy effect concentrated?

Figure 6 (Appendix B.3) summarizes three scope checks. Steering is strong at the first query but near zero at separately sampled states after three or six open-loop chunks; because these are different physical states, this is state dependence rather than causal temporal decay. The first-query effect is concentrated in the wrist-camera future (0.435), with smaller primary-camera steering (0.101) and no measurable proprioceptive steering (0.001). The paired donor also retains its largest advantage over self and Gaussian controls and a smaller but positive advantage over natural and shuffled futures. Thus, the Cosmos Policy result is localized to early visual futures rather than a generic response to any replacement.

4 Stage 2: K/V Pathway Patching

Stage 1 asks whether inserting a different predicted future changes behavior in a specific direction. Stage 2 asks how information from that inserted future reaches the action. It is a separate, nested experiment: the donor future remains inserted while one internal attention route is changed.

4.1 Method

An attention layer can be understood as a lookup. An action token produces a *query*; comparing that query with each token’s *key* determines how strongly the action attends to that token. The token’s *value* is the information then mixed into the action representation. Future-token keys and values therefore determine both which future information the action reads and what information it receives.

The Stage-2 procedure has five steps:

1. **Save the original-future pathway.** Run the recipient with its own predicted future and record the future tokens’ keys and values.
2. **Reproduce the Stage-1 effect.** Insert the donor future into the recipient and verify that the recomputed action moves toward the donor action. This is the unpatched donor-future run.
3. **Replace only K/V.** Keep the donor future inserted, but replace its future-token keys and values with the states saved from the recipient’s original-future run.
4. **Recompute with everything else unchanged.** Preserve the observation, instruction, noise, token positions, and all non-targeted states, then recompute the action.
5. **Measure how much steering disappears.** Subtract the patched donor steering from the unpatched donor steering. We call this decrease *K/V replacement loss*. A large loss means that the replaced future-token K/V route carried much of the donor-specific influence; residual steering may use other or interacting routes.

4.1.1 Cosmos Policy: future-key removal

Action queries can attend to future keys. We continuously gate future keys at prespecified block 27, with exact all-key recomputation, equal-count random-key and current-key controls, and a block-0 control. This removal changes attention normalization, so it tests whether a late future-to-action edge is active, not the information carried by a token-count-preserving route.

Table 3: K/V pathway test across 22 states from six tasks. Replacement loss is the decrease in donor steering after future tokens supply recipient self-future K/V; intervals are 95% state-bootstrap intervals.

Outcome	Donor steering	Replacement loss [95% CI]	Relative reduction
Predicted action	0.992	0.877 [0.842, 0.910]	88%
Executed action	0.983	0.814 [0.723, 0.900]	83%
Executed endpoint	1.003	0.851 [0.766, 0.922]	85%

4.1.2 Cosmos 3: future-token K/V replacement

For Cosmos 3, the original-future cache records future-video keys and values at all 36 layers and all eight model forwards. During the donor-future run, replacement is replayed call by call at every layer, and only the action-query rows are recomputed. Text, current-video and action K/V, token count and order, observation, instruction, noise, and all other query outputs remain native to the donor-future run. The intervention therefore makes the direct future-to-action K/V route recipient-like while leaving the donor future itself present. Exact cache alignment and the separate layer-16 calibration are detailed in Appendix A.3. This is an operational pathway intervention, not a complete circuit decomposition.

Stage-2 statistical procedures and their relationship to Stage 1 are detailed in Appendix A.7.

4.2 Results

Future-token attention provides an active route from generated futures to action. Population-level K/V replacement in Cosmos 3 gives the stronger of the two pathway tests.

4.2.1 Cosmos Policy

Block-27 future-key removal produces a monotonic action-disruption curve: normalized L_2 change is 0.022, 0.035, 0.054, and 0.068 at gates 0.25, 0.50, 0.75, and 1.00. All-key recomputation is bitwise exact. Full removal is 0.016 more disruptive than equal-count random-key removal and 0.048 more disruptive than block-0 future-key removal; equal-count current-key removal is 0.024 larger. Executed endpoints support the same bounded conclusion, although state counts and intervals are not available for this analysis. The late future-to-action edge is active, but it is neither the only nor the largest route through the block.

4.2.2 Cosmos 3

The interface was selected on excluded calibration tasks and frozen before population testing. The results follow the logic of the intervention directly (Table 3; Appendix Figure 8). For predicted actions, mean donor steering falls from 0.992 before patching to 0.115 after replacing future-token K/V, a loss of 0.877, or 88%. Executed-action steering falls from 0.983 to 0.169, a loss of 0.814, or 83%. Executed-endpoint steering falls from 1.003 to 0.152, a loss of 0.851, or 85%. The loss is positive in all 22 states. Because the donor future remains inserted while token count, attention positions, and all non-targeted states remain fixed, this drop shows that donor-specific information travels through the future-token K/V interface. The nonzero residuals show that this interface carries most, but not all, of the effect. Task-balanced loss estimates are 0.877, 0.812, and 0.853; all prespecified primary sign tests remain significant after Holm correction.

4.3 Interpretation

Stage 2 localizes part of the Stage-1 behavioral effect without redefining the recipient-donor intervention. In Cosmos 3, future-token K/V carries most of the donor-directed shift, while the residual effect leaves room for additional or interacting routes. Cosmos Policy provides weaker but convergent evidence for an active late future-to-action edge. These results identify an influential pathway; they do not recover a complete circuit, exclude interactions with unpatched components, or establish planning.

5 Discussion and Limitations

Across both WAM generations, replacing the recipient’s predicted future with a coherent donor future predicts the direction of the resulting action and simulator endpoint. RIFT established dependence on an organized future interface; our result adds content-specific sufficiency by using a policy-native alternative to predict the sign of behavioral change. The checkpoints nevertheless use different content: Cosmos Policy’s early-state pattern resembles inverse dynamics from visible robot motion, whereas Cosmos 3’s whole-future effect is not recreated by isolated robot or object pixels. Its useful signal may be distributed, contextual, or nonlinear. Pathway tests further locate an active late edge in Cosmos Policy and show that future-token K/V carries most of Cosmos 3’s donor-directed shift.

This evidence is narrower than planning, which would require generating alternatives, predicting their outcomes, comparing them using a goal or value, and selecting accordingly. We do not test that sequence, and no screened pair retained a stable success/failure label across continuation seeds. Here, “future” and “imagined” describe generated representations, not deliberation: these direct policies condition action on coherent futures, but need not plan over consequences.

Limitations. All interventions are in simulation and cover two checkpoints, six RoboLab tasks, and LIBERO-Long. Cosmos Policy’s strongest result uses one first-query state per task; later estimates use different states, its natural content analysis has only four states, and rendered hybrids may be off distribution. Cosmos 3 completed 22 of 24 frozen population states and 10 of 12 factor states. Pixel masks may miss distributed features, key removal changes attention normalization, and K/V replacement can interact with unpatched routes. We neither demonstrate task-success mediation nor recover a full circuit.

Broader impact. Mechanistic tests can distinguish plausible predictions from representations that control behavior, but the same interfaces could redirect actions or create false confidence on narrow state distributions. Use beyond simulation requires broader coverage, physical safety constraints, and monitoring outside the intervention metric.

6 Conclusion

Inserting one normal rollout’s future into another turns “future use” into a directional causal test. Cosmos 3 actions and executed endpoints move almost completely toward the inserted alternative across 22 states and six tasks, and future-video K/V replacement removes 83–88% of that steering. Cosmos Policy shows a smaller early-state effect that survives execution and exceeds natural, shuffled, and geometry-matched controls. Together, these results show that generated futures are not merely correlated auxiliary outputs: their specific content can help determine what the policy does, through an identifiable but incomplete internal route.

The next step is to move from causal conditioning to consequence-sensitive choice. Experiments should hold predicted robot motion fixed while varying task outcomes, test whether goal or value representations mediate the response, and measure closed-loop success rather than one-chunk endpoints. Finer head- and layer-level patching can resolve the residual Cosmos 3 pathway, while broader checkpoints, tasks, and physical robots can test generality. That progression would separate inverse-dynamics-like use of a future from genuine comparison among futures—the missing evidence needed to turn a claim about future-conditioned action into a claim about planning.

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A Reproducibility and Protocol Details

A.1 Experimental settings and frozen designs

The Cosmos Policy experiments use the public `Cosmos-Policy-LIBERO-Predict2-2B` checkpoint, `LIBERO-Long`, `PyTorch 2.7` with `CUDA 12.8`, and two `NVIDIA H100 80GB` GPUs. The frozen design contains ten tasks, three state indices, eight native policy runs per state, one reciprocal donor pair per state, and three future-noise draws per direction. The observation prefix, instruction, recipient action noise, future-noise schedule, tokenizer settings, and denoising schedule are held fixed within an intervention comparison.

The Cosmos 3 experiments use the public `16B Cosmos3-Nano-Policy-DROID` checkpoint with the released Cosmos, Cosmos Framework, and RoboLab evaluation stack. The frozen population screen selected 24 state clusters across six eligible tasks, capped at four clusters per task, using only native success, donor separation, and replay-validity information. Twenty-two clusters completed. The separate robot/object factor study selected 12 clusters by native object-position separation and completed its prespecified minimum of ten across all six tasks.

A.2 How the donor future is patched into the recipient run

Cosmos Policy. The donor target is a clean latent encoded with the public tokenizer from the RGB and proprioceptive endpoint reached by a normal 16-action rollout. In the released latent layout, temporal indices 5, 6, and 7 are future proprioception, wrist RGB, and primary RGB. At initialization and at each of the five solver evaluations ($\sigma = 80.0, 42.291, 20.972, 9.618, 4.0$), we replace only these recipient future coordinates with the donor values. The denoiser input at noise level σ is $F_D + \sigma \epsilon_R$, where ϵ_R is the recipient’s fixed future-noise draw, and the returned clean- x_0 estimate is reset to F_D at the same coordinates. The action and value coordinates are never written. Thus, the donor future is reimposed at every solver evaluation rather than inserted once into

a noisy state or predicted velocity. The evaluated direct-policy configuration is always conditional (its unconditioned input is `None`), so every solver call uses the wrapped x_0 function.

Cosmos 3. The prepared rectified-flow state is flattened as vision then action: vision has shape $[1, 48, 9, 33, 40]$ and action $[33, 64]$. We preserve latent frame 0 and replace future-video frames 1–8 with the donor target throughout denoising. A target F_D is either the final clean video latent from a normal generation or the public-tokenizer encoding of a 33-frame video produced by executing a normal 32-action rollout. Before both model forwards at each of four UniPC calls, future vision is set to $(1 - \sigma)F_D + \sigma\epsilon_R$. After guidance-3 combines the predicted velocities, we replace only future-vision velocity by $(x_\sigma - F_D)/\sigma$, making its clean estimate F_D at the sampler boundary. The recorded sigmas are 0.999, 0.937, 0.833, 0.624. The current-video and action state and action velocity are never changed.

A.3 K/V pathway implementation

The Cosmos 3 self-future run records future-video keys and values separately for all 36 layers and all eight model forwards: four denoising calls, each with conditional and unconditional passes. Cache entries are replayed call by call in the donor-future run, and only action-query rows are recomputed. Current-video and future-video query outputs remain native; text, current-video, and action K/V are unchanged. A calibration scan on excluded tasks selected layer 16 for single-layer localization, whereas the population estimate replaces all 36 direct future-K/V-to-action-query interfaces.

A.4 State restoration, execution, and missingness

Each endpoint comparison begins from an identical saved simulator state. Evaluation reconstructs the environment, restores that state, replays the observation prefix, verifies the restored observation, and then executes one recomputed action chunk. Recipient and donor endpoints are produced by the same restoration and execution procedure. This design makes the endpoint projection a comparison between reachable continuations of one physical state rather than between unrelated trajectories.

Two frozen Cosmos 3 population units were unavailable because their registered source branches ended after 21 and 29 actions, before producing the required 32-action, 33-frame intervention source. Neither unit was replaced, and no intervention outcome entered the decision to retain or omit a state. The completed population therefore exceeds the prespecified minimum of 20 states across at least five tasks. The factor study likewise meets its minimum of ten states across at least five tasks.

A.5 Intervention validity and controls

The self-future condition passes the recipient’s own future through the complete intervention machinery. For Cosmos 3, identity token-layout capture, sampler wrapping, cache record/replay, and patching the recipient future back into its own run reproduce baseline actions bitwise. Sentinels at the sampler boundary confirm that replacing recipient future coordinates with donor values does not write directly into action coordinates. The future-token pathway experiment preserves token count and order while leaving current-video, language, action, and all non-targeted key/value states unchanged.

The geometry-matched Gaussian control has the same norm and recipient distance as the donor displacement but no coherent content. Natural controls are valid unpaired continuations selected without intervention outcomes, and shuffled controls use another future from the same task. These comparisons distinguish exact recomputation, perturbation magnitude, generic alternative-future effects, and pair-specific content. Executed-future conditions add a separate realizability check by encoding frames generated when a native donor action is physically replayed in the simulator.

A.6 Stage-1 statistical analysis

The independent unit is a restored simulator state, not an individual noise draw or recipient–donor direction. Reciprocal directions, donor choices, noise draws, and repeated processes are averaged within state before any population summary. Reported 95% intervals use 10,000 percentile-bootstrap resamples of states [4]. For the common-scale condition estimates in Table 1, each control’s projection is reconstructed within unit from the registered pairwise contrasts before bootstrapping; these

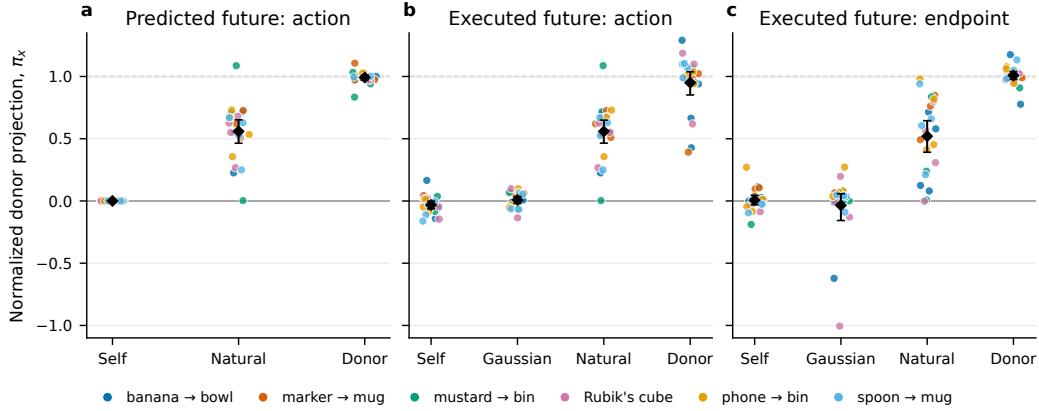


Figure 4: Cosmos 3 condition-level projections across 22 restored states for predicted actions, executed actions, and endpoints. Colors identify tasks; black diamonds and whiskers are means and 95% state-bootstrap intervals.

are not differences of aggregate means. Task-balanced sensitivity analyses first average states within task and then resample tasks.

We prespecified 0.10 projection units—10% of the native recipient-to-donor displacement—as the smallest effect of interest for the Cosmos Policy confirmatory family and the Cosmos 3 population and factor families [9]. An interval contained within $[-0.10, 0.10]$ supports practical equivalence to zero at that scale. Prespecified primary sign tests use state-level directions and Holm correction within each family.

A.7 Stage-2 statistical analysis

Stage 2 reuses the Stage-1 independent-unit and bootstrap conventions. For Cosmos 3, replacement loss is formed within restored state by subtracting K/V-patched donor steering from the corresponding unpatched donor steering before population aggregation. Task-balanced sensitivity estimates and family-wise Holm correction follow the Stage-1 procedure. The Cosmos Policy key-removal study is interpreted descriptively because state counts and intervals are unavailable for that diagnostic.

B Additional Experiments and Diagnostics

B.1 Condition-level Cosmos 3 projections

Figure 4 shows the condition estimates before they are converted into donor-minus-control contrasts. Predicted self-future recomputation stays at zero, natural futures produce partial donor-directed movement, and paired donors cluster near one. For executed futures, the matched Gaussian remains near zero while both action and endpoint effects preserve the same ordering. The state-level display also makes clear that the endpoint conclusion is not produced by a single task or a single extreme unit.

B.2 Displacement-magnitude diagnostic

Directional projection could be misleading if a large normalized estimate arose from an extremely small native denominator or if patching the donor future caused only a negligible displacement. Figure 5 therefore compares unnormalized recipient-to-donor separation with the displacement caused by replacing the recipient’s future with the donor’s. Predicted-action and executed-endpoint displacements lie close to the identity line across the observed range; executed-action displacement is more variable but remains on the scale of the native difference. This diagnostic supports the interpretation that the intervention recreates a substantial behavioral displacement, not merely a tiny change that happens to align with the donor axis.

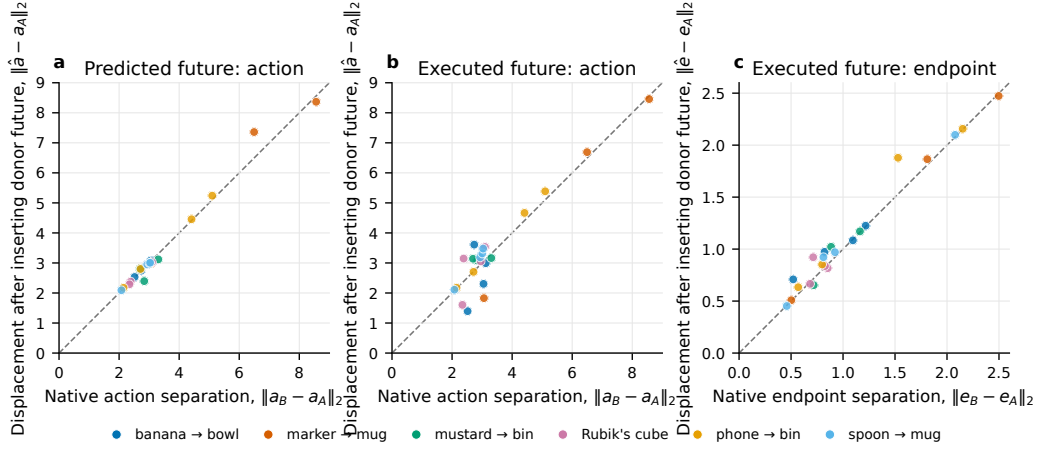


Figure 5: Native recipient-to-donor separation versus future-insertion displacement for Cosmos 3 predicted actions, executed actions, and endpoints. Dashed lines indicate equality; colors identify tasks.

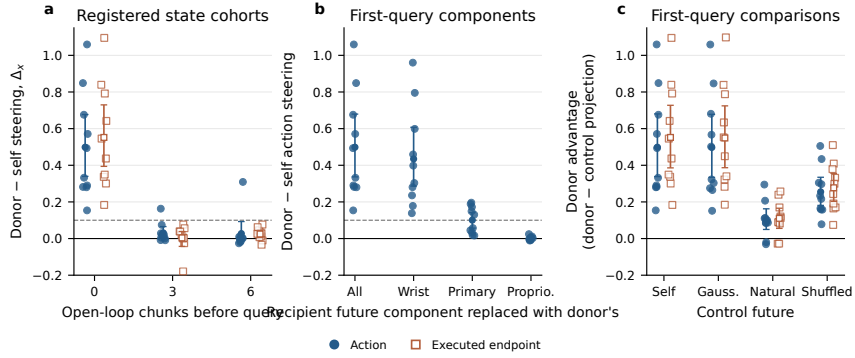


Figure 6: Cosmos Policy scope checks: (a) state cohorts, (b) future components, and (c) donor-minus-control contrasts. Small markers are tasks; large markers and whiskers are means and 95% task-bootstrap intervals.

522 B.3 Cosmos Policy timing and cross-domain checks

523 Across all 30 registered Cosmos Policy states, mean donor-minus-self steering is 0.186
 524 ($[0.095, 0.290]$) for actions and 0.194 ($[0.091, 0.305]$) for endpoints. The first-query, after-three-
 525 chunk, and after-six-chunk action estimates are 0.499, 0.031, and 0.027; the corresponding endpoint
 526 estimates are 0.552, 0.004, and 0.025. Because the three indices represent different physical states
 527 that may be nested within task trajectories, the first-query analysis is the cleaner confirmatory result
 528 and the differences should not be read as causal temporal decay.

529 The six-state RoboCasa replication is exploratory and is not pooled with LIBERO [13]. Donor-
 530 minus-self action steering is 0.0144 ($[0.0034, 0.0276]$), but its donor-minus-Gaussian contrast is only
 531 0.0025 and has an interval crossing zero. Endpoint steering is 0.0287 ($[0.0101, 0.0492]$), positive
 532 in 6/6 states; donor-minus-Gaussian endpoint steering is 0.0187 ($[0.0063, 0.0329]$), also positive
 533 in 6/6. The replication therefore supports a small executed-state effect while leaving action-space
 534 specificity uncertain.

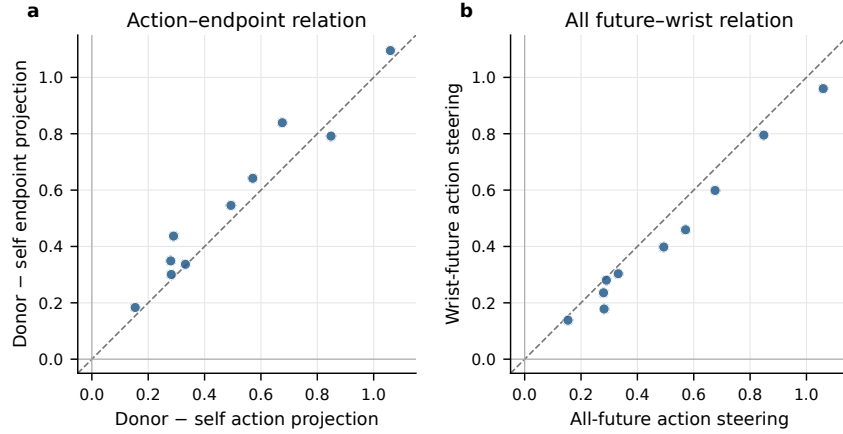


Figure 7: Cosmos Policy task-level relationships at the first query: (a) action versus endpoint steering and (b) all-future versus wrist-future steering. Points are tasks; dashed lines indicate equality.

535 B.4 Within-task relationships in Cosmos Policy

536 Figure 7 provides two task-level diagnostics at the first query. Action and executed-endpoint steering
 537 generally rise together, supporting the use of simulator execution as a behavioral continuation of the
 538 action result. Wrist-future steering also tracks all-future steering across tasks, consistent with the
 539 aggregate modality decomposition while showing that the wrist view does not reproduce the entire
 540 effect uniformly.

541 B.5 Additional content and pathway checks

542 The Cosmos Policy rendered 2×2 study supplies a manipulation check for the object-content null.
 543 Decoded futures identify the intended hybrid cell with 92.5% accuracy in the primary camera and
 544 98.75% in the wrist camera, while executed endpoints identify the four cells at the 25% chance
 545 rate. Thus, the future representation visibly contains the imposed distinction even though the pre-
 546 registered object main effect on action and goal-state endpoint is approximately zero. The com-
 547plementary natural-pair search found only four states with both an object-divergent/robot-matched
 548 and a robot-divergent/object-matched donor at a common recipient; these exploratory data favor the
 549 robot-motion donor, but the available population is below the planned minimum of ten.

550 The block-27 Cosmos Policy key-removal analysis includes two additional checks beyond the action-
 551 space dose curve reported in the main text. Full future-key removal changes the executed endpoint
 552 by an absolute mean projection of 0.281 ([0.164, 0.430]; 10/10 first-query states) and exceeds equal-
 553count random-key removal by 0.169 ([0.058, 0.322]; 9/10). Its difference from equal-count current-
 554key removal is -0.035 with an interval crossing zero. The endpoint analysis therefore supports an
 555active future route at the first query while reinforcing the conclusion that this route is not uniquely
 556dominant over current-state information.

557 For Cosmos 3, task-balanced pathway-loss estimates are 0.877 for predicted actions, 0.812 for
 558 executed-future actions, and 0.853 for executed endpoints, closely matching the state-weighted esti-
 559mates. The remaining steering after self-future K/V replacement is 0.115, 0.169, and 0.152, respec-
 560tively. This residual motivates finer-grained tests of parallel attention heads, non-attention routes,
 561and interactions between patched and unpatched components rather than a claim that the selected
 562interface is the complete mechanism.

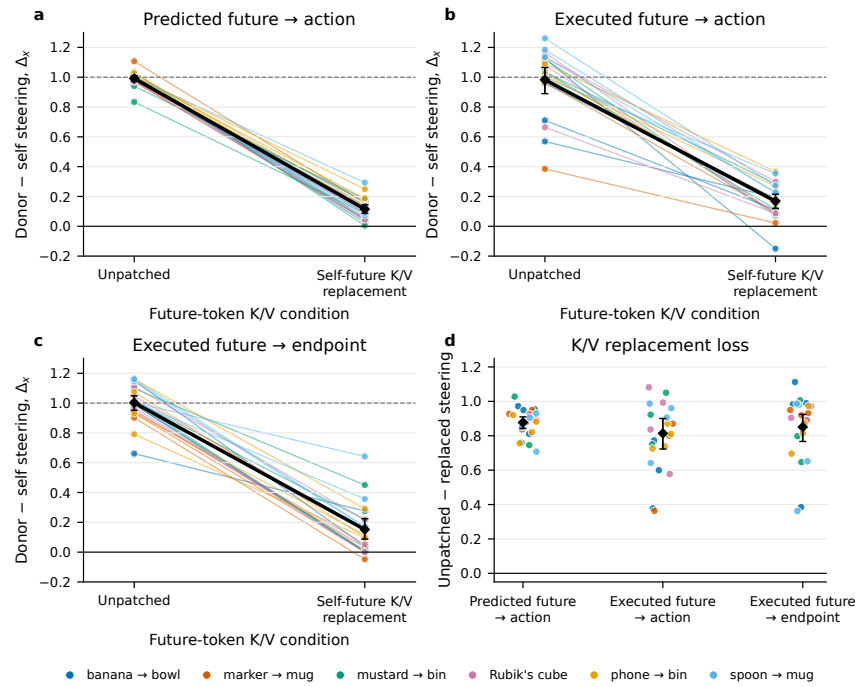


Figure 8: Cosmos 3 donor steering before and after future-token K/V replacement. Lines connect the same state; black diamonds and whiskers are means and 95% state-bootstrap intervals. Panel (d) shows replacement loss.